



Investigating the persistence of iron-bound phosphorus in lake sediment under sulfidic conditions

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Abstract Iron (Fe) additions are widely used to suppress internal phosphorus (P) loading in eutrophic lakes by promoting the formation of vivianite ($\text{Fe(II)}_3(\text{PO}_4)_2 \cdot 8\text{H}_2\text{O}$), a redox-stable Fe–P mineral that can sequester P in anoxic sediments over the long term. However, vivianite is unstable under sulfidic conditions, but the remobilization potential of P is unclear. In a 96-day mesocosm experiment, we examined how sulfate availability affects Fe-bound P persistence in a controlled model system of Fe-amended lake sediment. Sediment amended with Fe–P and enriched with vivianite during an anoxic pre-incubation was subsequently incubated under

oxic overlying water at either low ($<100 \mu\text{mol L}^{-1}$) or high ($\sim 1 \text{ mmol L}^{-1}$) sulfate concentrations. Pore-water profiles measured with microsensors and diffusive gradients in thin films (DGT), combined with solid-phase analyses by sequential extraction, scanning electron microscopy coupled to energy dispersive spectroscopy (SEM-EDS), and X-ray diffraction (XRD), revealed that vivianite persisted in sediment under oxic water conditions but was progressively destabilized as sulfide accumulated through microbial sulfate reduction. Sulfide drove coupled dissolution–reprecipitation reactions that transformed vivianite-bound Fe into amorphous Fe sulfides and pyrite. As a result, total sediment P-binding capacity declined. Although a portion of released P was re-adsorbed onto freshly precipitated Fe(III) phases in the oxic surface layer, this only partially compensated for the net loss of P-binding capacity, and total solid-phase P

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declined in both treatments. These findings highlight that P retention is linked to sulfide availability, and that monitoring porewater sulfide concentrations may serve as a practical early warning indicator of declining P retention capacity.

Keywords Vivianite · Internal phosphorus loading · Lake restoration · Sulfate reduction · Phosphorus mobilization · Iron-bound phosphorus

Introduction

Effective phosphorus (P) management is often necessary for the restoration and preservation of aquatic environments (Schindler 2006). Water quality targets, such as those defined by the European Water Framework Directive, aim to ensure that all water bodies reach "good" ecological status by 2027. The urgency for lake water quality improvement is therefore increasing, and freshwater managers are increasingly considering interventions targeting internal P loading — the net release of P from the sediment to the water resulting from decades of legacy P accumulation. Limiting internal P loading is therefore a priority, whether by reducing net P release from the sediment or by supporting the P-binding capacity within the sediment to accelerate removal from the water column (Tammeorg et al. 2024).

One important management strategy is enhancing P retention in the sediment by treatment with P-binding materials (Lürling et al. 2016), but it remains essential to evaluate the effectiveness, broader environmental impacts, and long-term persistence of these measures. Iron (Fe) compounds bind P efficiently in various forms, and Fe-based materials are cost-effective and readily available compared to other P-binding compounds such as lanthanum-modified clays (Phoslock®; Lürling et al. 2016). While materials such as aluminum may cause toxicity or other unwanted effects requiring extensive testing and monitoring (Lürling and Mucci 2020), the risks associated with Fe amendments are generally low (Cooke et al. 1993; Bakker et al. 2016), and its biogeochemistry is comparatively well understood given the ubiquitous role of Fe in the natural cycling of P in aquatic systems (Shaked et al. 2004; Maranger et al. 2006).

Despite these advantages, Fe amendment efforts have yielded mixed results, often showing a return to

elevated P concentrations after some time (Heinrich et al. 2022; Münch et al. 2024). The redox sensitivity of the Fe(III) compounds typically used for amendment, as well as the affinity of Fe towards competing ligands — particularly organic matter (OM) and sulfide (S^{2-}) — have been proposed as key factors limiting long-term success (Rothe et al. 2015; Heinrich et al. 2022). Eutrophic sediments exposed to anoxic bottom water undergo reductive dissolution of iron (oxyhydr)oxides (FeOx) and concomitant P release (Cooke et al. 1993; Hupfer and Lewandowski 2008). Kleeberg et al. (2013) proposed that sediment OM and sulfur (S) contents should be considered when calculating Fe doses to account for competing ligand demand but argued that re-precipitation of Fe and P at the oxycline restored P retention. However, while re-precipitation has resulted in long-term P removal from the epilimnion in some dimictic lakes after Fe amendment, this keeps a reactive P reservoir which returns bio-available following bottom water anoxic events, upon FeOx reductive dissolution. This redox-cycling thus limits the P retention of the sediment by FeOx. Consequently, long-term P immobilization depends on P being incorporated into redox-stable Fe(II) minerals, most notably vivianite ($Fe(II)_3(PO_4)_2 \cdot 8H_2O$) (Rothe et al. 2016). Vivianite is widely distributed in aquatic sediments and has been identified as an important sedimentary P sink (Egger et al. 2015; O'Connell et al. 2015; Dijkstra et al. 2018; Haasler et al. 2025). From a lake management perspective, its redox stability makes vivianite particularly attractive for long-term P storage, especially given its favorable molar Fe:P of 1.5, compared to FeOx, whose ratio typically declines below ~10 (Jensen et al. 1992; Gunnars et al. 2002; Kleeberg et al. 2012). Both laboratory and *in-situ* studies have demonstrated that vivianite can form within months in Fe-enriched sediment under anoxic conditions following Fe amendment (Heinrich et al. 2020; Kubeneck 2024).

However, vivianite is unstable in the presence of sulfide, as shown by exposing either synthetic vivianite (Kubeneck et al. 2024) or natural vivianite-rich sediments (van Kuppevelt et al. 2025) to sulfidic conditions. Under anoxic conditions, sulfide produced by microbial sulfate (SO_4^{2-}) reduction readily reacts with Fe via two main pathways: reductive dissolution of Fe(III)phases, releasing associated P to the overlying water and precipitating elemental sulfur (S^0)

(Dos Santos Afonso and Stumm 1992; Poulton et al. 2004), and direct reaction with Fe(II) (including vivianite) to form thermodynamically stable iron sulfide (FeS_x) phases, (Nriagu 1972; Ma et al. 2023). These processes can occur simultaneously and outcompete vivianite formation (Roden and Edmonds 1997; Rothe et al. 2015), and although Fe-rich sediments may retain some P under reducing conditions, microbial sulfate reduction can induce net P release from the solid phase (Chen et al. 2016; Zhao et al. 2021).

Despite this evidence, the mechanisms of P release following Fe amendment remain poorly constrained. Detailed investigations of vivianite persistence under well-defined sulfidic conditions are still lacking, limiting our ability to predict the long-term effectiveness of Fe-based restoration on tangible terms. In this study, we tested the hypothesis that vivianite stability decreases under elevated sulfidic conditions, using a novel approach with Fe-amended lake sediment that allows simultaneous investigation of Fe–P and Fe–S interactions in a model system employing layered sediment mesocosms. While it was previously thought that bottom water anoxic conditions

were prerequisite for P mobilization from Fe phases, we investigate the processes under oxic bottom water conditions typical for a shallow lake, where P mobilization takes place in the sulfidic zone directly below the oxic layer. This sulfidic zone was identified at high vertical resolution, in contrast to previous studies that lacked spatially resolved sulfide measurements. This enabled a quantitative assessment of the sulfide reaction front and the mechanisms by which sulfide availability affects vivianite stability and the sedimentary P pool.

Material and methods

Collected and homogenized sediment was used to compose layered cores of amended and unamended sediment, which were incubated in a controlled mesocosm experiment with contrasting sulfate availability (Fig. 1). Constructed cores from homogenized sediment were chosen as experimental units rather than intact cores, as this allowed control of the vivianite enrichment and a uniform starting composition

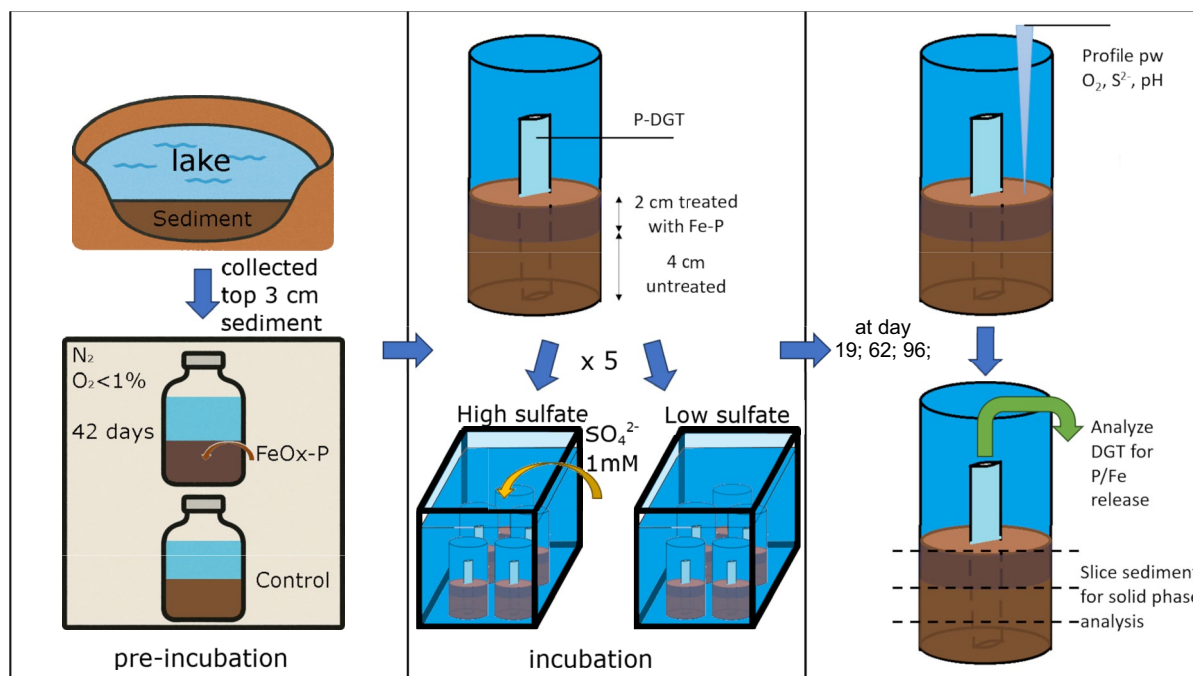


Fig. 1 Visual summary of the experimental procedure. Sediment was collected and pre-incubated under controlled conditions (left). Subsequent sulfate incubations were performed to

investigate redox and phosphorus dynamics (middle). Post-incubation analyses included microsensor profiling and sediment fractionation for geochemical analysis (right)

between replicates (Table 1). This homogeneous initial condition was essential to assess later variation due to the sulphate treatment and for correlating pore-water and mineral phase at high spatial resolution. After Fe amendment and pre-incubation of the sediment, replicate cores were constructed and randomly assigned to a high sulfate (1 mmol L^{-1}) or low sulfate (0 mmol L^{-1}) treatment, $n = 5$ per treatment. The first two replicate cores were removed and analyzed one per treatment at respectively 19 and 62 days after the start of the incubation to monitor the changing composition over time, while the remaining three replicates were evaluated together at the end of the experiment (96 days) to account for between-core variation.

Sediment collection and pre-incubation

Sediment was collected in May 2023 from Langer See (N52.24333°, E13.78805°), a shallow (max. depth 4 m), eutrophic freshwater lake in Eastern Germany characterized by low iron content and high primary productivity (water chemistry detailed in Table S1). As the focus of this study was the stability of vivianite following Fe amendment, Langer See was selected for the following reasons: First, vivianite enrichment following Fe addition is comparatively higher in its Fe-poor sediments than in more Fe-rich lakes (Heinrich et al. 2020). Secondly, the lake water is sulfate-rich ($1094 \pm 336 \text{ } \mu\text{mol L}^{-1}$, monthly measurements Jan. 2022 – May 2023, detailed in the supplementary material S1; within the range reported by Nixdorf et al. 2003) and the sediment has a high OM content available for dissimilation. Twenty-five

sediment cores were collected with a piston corer (UWITEC GmbH, Mondsee, Austria) of 6 cm internal diameter, deployed at 3.4 m water depth. Surface sediments from the top 3 cm were pooled to obtain approximately 2 L of homogenized sediment, processed in the laboratory on the day of sampling.

Prior to the main incubation, sediment was amended with iron and phosphate and pre-incubated under anoxic conditions to enrich it with vivianite, following Heinrich et al. (2020). Amorphous Fe hydroxide (FerroSorp® Plus, HeGoBiotec, Berlin, DE) was powdered and 10.1 g suspended in a solution of 9.73 g monosodium phosphate dihydrate ($\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$) in ultrapure water to achieve P adsorption. The suspension was mixed on an overhead shaker (Heidolph Scientific Products GmbH, Schwabach, DE) for 2 days, then centrifuged (9000 rpm, 5 min) and washed 9 times with ultrapure water. The resulting phosphate-loaded iron hydroxide (FeOx-P) was re-suspended in 125 mL ultrapure water, of which 70 mL was mixed into 1 L of lake sediment to increase the sedimentary Fe content to approximately three times the background value (Table 1). The Fe-P amended sediment was incubated in a closed 1 L bottle without headspace; an unamended bottle served as a control and was incubated in parallel. Both bottles were stored in air-tight jars with oxygen absorption sachets in a nitrogen (N_2)-flushed glovebox ($\text{O}_2 < 1\%$; Labstar, M.Braun Inertgas-Systeme GmbH, Garching, DE). The sediment was periodically mixed and sampled under N_2 atmosphere for sequential extraction to monitor FeOx-P conversion to vivianite. After 42 days, final samples were taken and both sediment types were processed for the subsequent incubation.

Table 1 Composition of the sediment after 42 days of anoxic (pre-)incubation when amended with Fe-P compared to the sediment without Fe amendment. The procedure followed that of Heinrich et al. (2020), and similar results were produced, with variation within less than an order of magnitude

		Measured value: this study/Reference: Heinrich et al. (2020)			
		Fe-P amended		Control	
		measured	reference	measured	reference
Concentration	Fe ($\mu\text{mol g}^{-1}$ DW)	444	631 ± 8	179	219 ± 4
	P ($\mu\text{mol g}^{-1}$ DW)	245	319 ± 2	40	62 ± 9
	S ($\mu\text{mol g}^{-1}$ DW)	604	476 ± 7	584	508 ± 13
	LOI (wt%)	27 ± 1	23 ± 1	28 ± 1	24 ± 1
Molar ratio	Fe:P	1.8	1.98 ± 0.02	4.5	3.63 ± 0.50
	S:Fe	1.4	0.75 ± 0.01	3.3	2.32 ± 0.02
	BD-Fe:BD-TP	1.53 ± 0.04	1.64 ± 0.12	1.21 ± 0.02	2.22 ± 0.26
Transformed BD-P → NaOH-P (%)		52	72		

Vivianite stability under different sulfidic levels

The pre-amended sediment and the control were incubated in aerated water under high and low sulfate treatments to investigate the influence of sulfate concentration on Fe mineral composition (Fig. 1). Both sediment types were homogenized and a subsample preserved at -20°C , then distributed in ten plexiglass® tubes (five replicates per treatment, 6 cm diameter, 10 cm height) as follows: first, 4 cm of unamended sediment was added above an adjustable rubber stopper as an internal control, then 2 cm of Fe-P amended sediment was carefully layered on top, and the tube was slowly filled with water to avoid resuspension.

The cores were placed in two identical 20 L aquaria (five cores per treatment) filled with 15 L of artificial lake water (Fig. 1), prepared following Smith et al. (2002), and adapted to match the mean chemical composition of Langer See surface water (Table S1, $n = 19$), but without adding sulfate. The water was made from a 1000× dilution of a stock solution of 81.386 g magnesium chloride hexahydrate ($\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$), 125.030 g calcium chloride

dihydrate ($\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$), 20.150 g potassium bicarbonate (KHCO_3), and 120.111 g sodium bicarbonate (NaHCO_3) in 1 L Millipore water. This ensured stable, homogeneous water composition across treatments and a sulfate concentration of $6.8 \pm 0.3 \mu\text{mol L}^{-1}$ in the low sulfate aquarium at the start of the experiment. For the high sulfate treatment, 150 mL of 0.1 mol L^{-1} sodium sulfate (Na_2SO_4) was added to target a concentration of 1 mmol L^{-1} , corresponding to the average sulfate concentration in Langer See. Sulfate concentrations were confirmed by ion chromatography (IC; Dionex Ionchromatograph system 2500, Thermo Fisher Scientific Inc., Waltham, MA, USA) after filling and monitored periodically throughout the experiment.

The overlying water was continuously aerated to maintain oxic conditions and ensure homogeneous water chemistry without disturbing the sediment surface. After one day of settling, the sediment surface was carefully lifted by pushing up the adjustable stopper up to 0.5 cm below the rim on the tubes (Fig. 2a,b), improving water circulation and access for microsensor profiling. On day 15, diffusive gradients in thin films (DGT) probes (LSPX-NP, DGT®

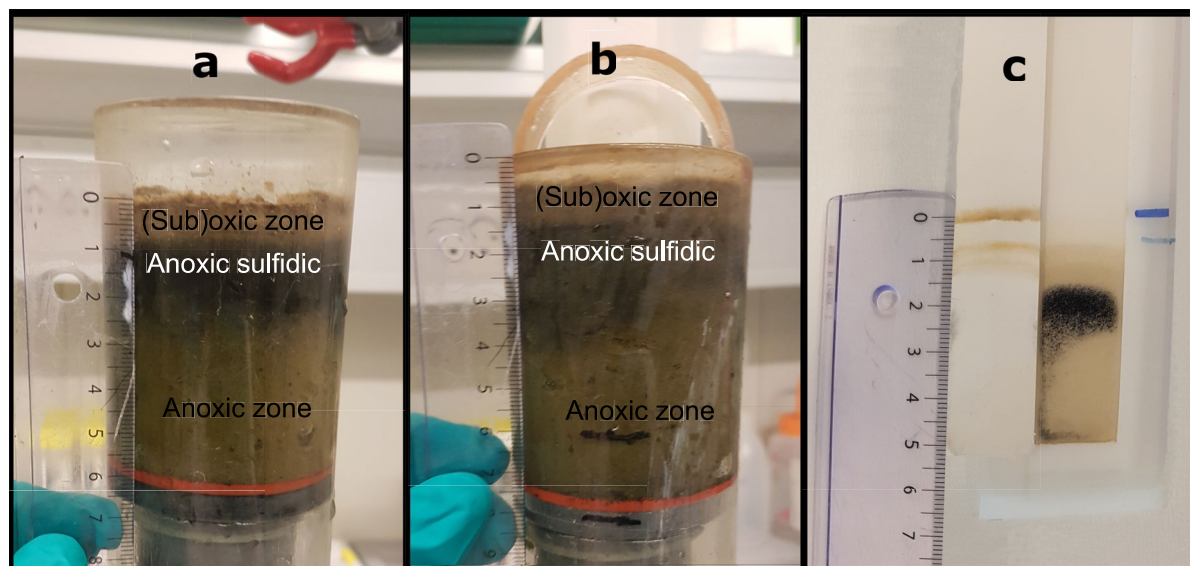


Fig. 2 a) and b) Layered sediment after 96 days of exposure to high and low sulfate treatments, respectively. b: Below the water column there is a light-colored layer which includes the upper oxic zone, and a suboxic zone directly below. Black coloration marks the sulfidic zone, where sulfide is produced and reacts with Fe which precipitates as black iron sulfide miner-

als. The lighter colored sediment below is also anoxic, but the lack of available Fe prevents black precipitation. c) The DGT gel on day 96 after 81 days of exposure in the high sulfate treatment. Black precipitation is visible in the gel at the same depth as in the sediment, and orange-brown Fe(III) precipitation forms a layer near the SWI

Research, Lancaster, UK) were inserted into four replicate cores per treatment, leaving one core per treatment undisturbed. Cores were incubated for 96 days at 20 °C in darkness to prevent primary production. Porewater parameters (oxygen and sulfide concentrations; pH) were monitored by microsenors throughout the duration of the experiment. One replicate core with DGT from each treatment was sacrificed for porewater and solid-phase analysis after 19 and 62 days; the remaining three cores per treatment were processed at the end of the experiment (day 96).

Microsensor and DGT profiling

Porewater depth profiles of dissolved O₂ (DO), hydrogen sulfide (H₂S), and pH were measured using custom-made (Revsbech and Jørgensen 1986; Revsbech 1989) or commercial microsensors (SULF sensor, Unisense, Aarhus, DK), calibrated following standard protocols as described in Marzocchi et al. (2018). Signals were recorded using a multimeter and SensorTraceSuite software (Unisense, Aarhus, DK). Sensor tips were positioned at the sediment-water interface (SWI) using a stereomicroscope and profiles measured from 1 cm above to 4 cm below the SWI at 250 µm steps using a computer-controlled micromanipulator (Unisense, Aarhus, DK), with 2 s waiting and measuring time at each depth. Sensors were moved approximately 1 cm laterally between profiles within the same core. Profiles were measured after one day of incubation and weekly thereafter until day 19, and prior to core retrieval at days 19, 62, and 96.

Diffusive gradient in thinfilm probes (LSPX-NP, DGT® Research, Lancaster, UK) captured porewater dissolved Fe(II) and P via accumulation in a mixed chelex/titanium oxide binding layer, handled and processed following the manufacturer guidelines. After retrieval, gels were rinsed and sliced at 0.5 cm intervals. Fe(II) was extracted with 1 mol L⁻¹ nitric acid (HNO₃) and P with 1 mol L⁻¹ sodium hydroxide (NaOH). Phosphorus was analyzed as soluble reactive phosphorus (SRP) photometrically at 880 nm using the molybdenum blue method (Murphy and Riley 1962), in a microplate reader. Iron was analyzed using inductively coupled plasma-optical emission spectroscopy (ICP-OES; iCAP 7000 series, Thermo Fisher Scientific Inc., Waltham, MA, USA).

Solid phase sampling and analysis

At each sampling time, one core per treatment was sliced at 1.2 cm intervals, yielding 4 slices per core. Sediment was homogenized, collected in 50 mL falcon tubes, and stored at -20 °C for up to one month before processing. For sulfide solid-phase analysis, sub-cores taken at the end of the experiment from the two cores without DGT were sliced at 0.5 cm intervals and frozen. Sediment was thawed before subsampling for dry weight (DW, determined after drying >12 h at 105 °C) and sequential P extraction. Samples were immediately refrozen after wet-phase analysis. Although freezing is not standard practice for sequential extraction, it was necessary for logistical reasons and has minimal effect on composition compared to alternative preservation methods (Rapin et al. 1986; Schlichting and Leinweber 2002). The remaining sediment was freeze-dried prior to X-ray diffraction (XRD) and scanning electron microscopy (SEM) analysis. Total P, Fe, S, and other elements including magnesium (Mg), manganese (Mn), and calcium (Ca) concentrations were determined by digesting freeze-dried sediment (50–100 mg) in a 1:3 (v/v) mixture of concentrated hydrochloric acid (HCl) and HNO₃ (DIN 38 414-S7 1983) and analyzing the digestate by ICP-OES (iCAP 7000 series, Thermo Fisher Scientific Inc., Waltham, MA, USA).

Phosphorus and sulfur sequential extractions

P pools in wet sediment were analyzed using the first three steps of the method by Psenner et al. (1984), modified by Hupfer et al. (1995). The Fe-P amended sediment was analyzed in triplicate periodically during pre-incubation and immediately before filling the cores. Homogenized slices from both treatments at the end of the incubation were analyzed in duplicate. Samples were thawed overnight at 8 °C in closed containers. This is a modification from the standard protocol, which is generally applied to fresh sediment, but was necessary for logistical reasons. Aliquots of 0.5 g were weighed into 15 mL centrifuge tubes and subjected to sequential extraction steps (10 mL each), separating the sediment by centrifugation (2500 rpm, 5 min) and decanting the supernatant at each step. Porewater P and loosely adsorbed P were extracted for 1 h with N₂-saturated ultrapure water (H₂O-P);

redox-sensitive P was released over 1 h with 0.11 mol L⁻¹ bicarbonate/dithionite (BD-P); and redox-stable P forms were extracted overnight with 1 mol L⁻¹ NaOH (NaOH-P). Total P (TP) in all fractions was determined after wet oxidation of 4 mL extract with 1 mL of 5 wt% K₂S₆O₈ for 2 h at 120 °C, then quantified as SRP at 880 nm using the molybdenum blue method (Murphy and Riley 1962). The NaOH-P fraction was further subdivided into inorganic mineral-bound P including stable Fe oxides and vivianite (NaOH-SRP, measured directly) and organic P (NaOH-NRP, calculated as NaOH-TP minus NaOH-SRP). Residual P was determined as the difference between the sum of extracted fractions and total P by ICP-OES. Iron concentrations in the BD and NaOH extracts (BD-Fe, NaOH-Fe) were measured by atomic absorption spectroscopy (AAS; pinAAcle 9001, PerkinElmer, Shelton, CT, USA).

Acid-volatile sulfide (AVS) and chromium-reducible sulfur (CRS) were quantified via cold distillation following Røy et al. (2014). Sediment samples (0.5–1 g wet weight) were transferred to reaction flasks with 20 mL dimethyl sulfoxide and purged with N₂ for 10 min. A 6 mol L⁻¹ HCl solution (8 mL) was injected to liberate AVS as H₂S, trapped in 5% zinc acetate (ZnAc) for 1 h. Subsequently, 16 mL of freshly prepared 1 mol L⁻¹ chromium(II) chloride (CrCl₂) in 2 mol L⁻¹ HCl was added under continuous N₂ flow to reduce CRS to H₂S, collected over 2 h. The CrCl₂ solution was prepared by reducing chromium(III) chloride CrCl₃ with zinc pellets under N₂ (indicated by a color change from green to blue). All traps contained silicone antifoam and citrate buffer (0.1 mol L⁻¹, pH 4). Sulfide in Zn traps was quantified using the Cline colorimetric method (Cline 1969). All procedures were conducted under N₂ atmosphere with continuous magnetic stirring.

Mineral identification

Scanning electron microscopy coupled to energy dispersive spectroscopy (SEM-EDS) was used to qualitatively investigate mineral phases in sediment collected at 0, 19, and 96 days from the 1.2–2.4 cm depth interval — below the oxic zone and spanning the different sulfide levels of the two treatments. Freeze-dried sediment (<1 mg) was deposited on adhesive carbon tape to form a thin layer and analyzed on a Zeiss LEO

435 VP SEM (Carl Zeiss AG, Oberkochen, DE) at 20 kV. Conduction was achieved with water vapor at low vacuum mode (Stokes 2008). Secondary electron imaging used a Large Field Detector; elemental composition was mapped by EDS using an Element-C2B detector (512 × 400 pixels; EDAX Team™ software, AMETEK, Inc., Berwyn, PA, USA). Phase identification used visual inspection and the built-in phase identification tool; elemental stoichiometry was calculated with the eZAF Smart Quant tool from combined EDS spectra.

Powder XRD was used to qualitatively investigate mineral composition, particularly vivianite, following Rothe et al. (2016). Analyses used a Bruker D2 Phaser with a Co X-ray source and SSD160 detector (Bruker Corporation, Billerica, MA, USA), measuring from 10 to 90° 2θ with a step size of 0.014° and 1 h per sample. The instrument was configured with a rotating sample holder, 1 mm fixed divergence slit, 2.5° primary and secondary Soler collimators, a fixed scatter screen at 3 mm, and an Fe Kβ filter. Mineral identification used Bruker DIFFRAC.EVA software with reference patterns from the Crystallography Open Database (Grazulis et al. 2009). For selected phases, the peak area under the largest non-overlapping reflection was calculated for semi-quantitative comparison between samples.

Data analysis

Total dissolved sulfide (TS) porewater concentrations were calculated from H₂S and pH profiles using the dissociation constant. Oxygen consumption rates were derived from microsensor profiles using the finite-difference method of Berg et al. (1998); parameters and assumptions are detailed in Table S2. The DGT-measured Fe(II) and SRP concentrations were used to calculate accumulation per resin surface area and mean porewater flux following the protocol of DGT Research Ltd (2020), with an elution correction factor of 0.9 for both analytes (Panther et al. 2014). Compositional variation between treatments and timepoints were tested for the depth averaged concentration of each duplicate, as well as for the individual depths. For each parameter, differences in concentration were assessed by two-way analysis of variance (ANOVA), with time and treatment as fixed factors, including their interaction, and the effects were tested

independently using Type II sums of squares (car package, v.3.1.3; Fox and Weisberg 2019). Correlations were tested by Pearson's correlation coefficient. Significance was set at $p < 0.05$. All statistical analyses and figures were produced in R version 4.4.1; additional figures were created in GIMP version 2.10.

Results

The collected sediment was low in P and Fe and high in OM (Table 1). Following the Fe-P amendment procedure of Heinrich et al. (2020), sediment P and Fe content increased by 500% and 150%, respectively, consistent with the enrichment reported in the original study (Table 1), confirming the reproducibility of the method (detailed comparison in supplementary material). A Fe:P of 1.6 ± 0.4 was determined in the P-loaded Fe-hydroxide slurry before addition to the sediment, and the same ratio was found in the BD fraction of the amended sediment during pre-incubation, translating to an overall Fe:P of 1.8 at the end of pre-incubation. During pre-incubation, P shifted from the BD-P fraction to the NaOH-SRP fraction (Fig. S1), consistent with the transformation of FeOx-P to vivianite (Heinrich et al. 2020). After 42 days, 52% of the initial BD-TP pool had been converted to NaOH-SRP. Since both BD-P and NaOH-SRP were major fractions at the start of the oxic incubation, the sediment likely contained both freshly formed vivianite and remaining, more recalcitrant FeOx phases.

Porewater conditions during sulfate treatment

The low sulfate treatment had an initial sulfate concentration of $6.8 \pm 0.3 \mu\text{mol L}^{-1}$, which increased over the incubation at a rate of $0.67 \pm 0.04 \mu\text{mol L}^{-1} \text{ day}^{-1}$ ($R^2 = 0.97$) but remained below $100 \mu\text{mol L}^{-1}$, typical for sulfate-poor lakes (Zak et al. 2021). The high sulfate concentration remained stable (ANOVA, $p = 0.167$) at $1046 \pm 46 \mu\text{mol L}^{-1}$ throughout.

Within one day of the start of incubation, an oxygen gradient formed in the top sediment (Fig. S2) and remained stable throughout the experiment: oxygen penetration depth (Table S3) — defined as the depth below which DO fell below 1% — did not differ between timepoints (ANOVA, $p = 0.2$) or between treatments (ANOVA, $p = 0.216$), which

showed similar concentration profiles (Fig. 3a). Oxygen consumption rates likewise did not change over time (ANOVA, $p = 0.441$) or between treatments (ANOVA, $p = 0.77$), averaging $19 \pm 5 \text{ mmol m}^{-2} \text{ d}^{-1}$.

After 19 days, higher porewater TS levels were found in the high sulfate treatment from 1 cm to at least 3 cm depth. At subsequent timepoints, TS increased progressively in the high sulfate treatment while remaining consistently low ($<50 \mu\text{mol L}^{-1}$) in the low sulfate treatment (Fig. 3b). In the high sulfate treatment, TS was highest at depth, decreasing towards the SWI and reaching zero at approximately 0.5 cm below it.

The DGT profiles of Fe(II) below the SWI differed markedly between treatments at the end of the incubation (Fig. 3c). The mean Fe(II) flux over the deployment period was approximately $6 \times 10^{-2} \text{ mmol m}^{-2} \text{ d}^{-1}$ in the high sulfate treatment and nearly twice as high ($\sim 0.11 \text{ mmol m}^{-2} \text{ d}^{-1}$) in the low sulfate treatment. Iron accumulation was lower in the high sulfate treatment (ANOVA, $p = 0.04$). The highest pronounced divergence was found in the sulfidic zone. Concentrations peaked at 1.5 cm depth in the high sulfate treatment (890 nmol cm^{-2}) and at 2.0 cm in the low sulfate treatment ($1160 \text{ nmol cm}^{-2}$).

The SRP uptake by DGT followed a similar pattern in both treatments. At day 19 (after 4 days of DGT deployment), SRP accumulation increased with depth from 0–2 nmol cm^{-2} above the SWI to 6–8 nmol cm^{-2} at 4 cm depth (Fig. 3d), corresponding to a mean flux of $1\text{--}2 \times 10^{-2} \text{ mmol m}^{-2} \text{ d}^{-1}$. By the end of the incubation (81 days of exposure), SRP accumulation had increased to 200–300 nmol cm^{-2} at all depths below the SWI, corresponding to a mean flux of $2\text{--}3 \times 10^{-2} \text{ mmol m}^{-2} \text{ d}^{-1}$. Detectable SRP above the SWI indicated net benthic P release, estimated at $<0.1 \text{ mmol m}^{-2} \text{ d}^{-1}$ across all treatments.

Sediment composition

Layering of Fe-P amended and unamended sediment produced a concentration gradient in the cores, with the highest Fe and P concentrations at the top and lowest at the bottom (Fig. S3). The overall composition of Fe, Ca, Mg and Mn was not different between treatments (ANOVA, $p > 0.05$).

Comparing day 96 with day 19, the overall P content decreased over time in both treatments (ANOVA, p

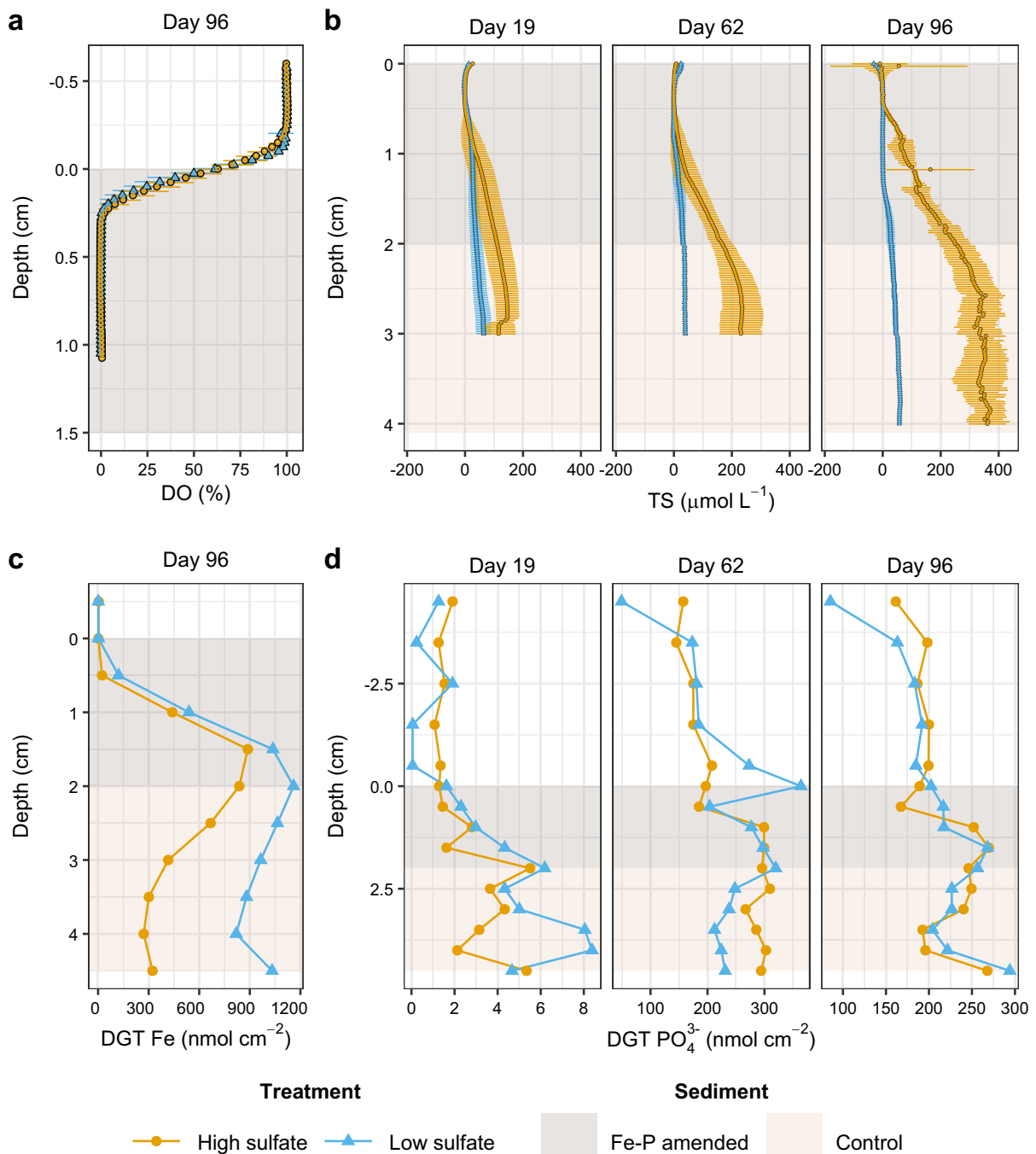
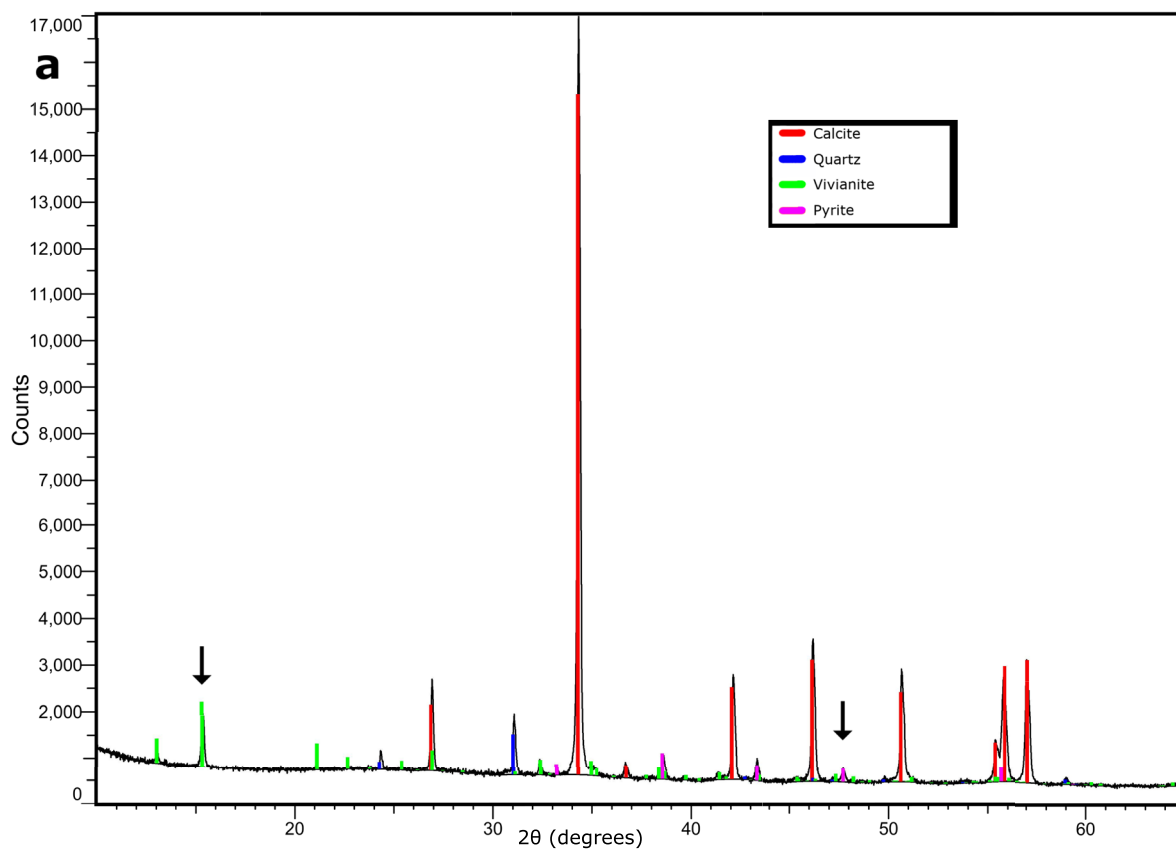
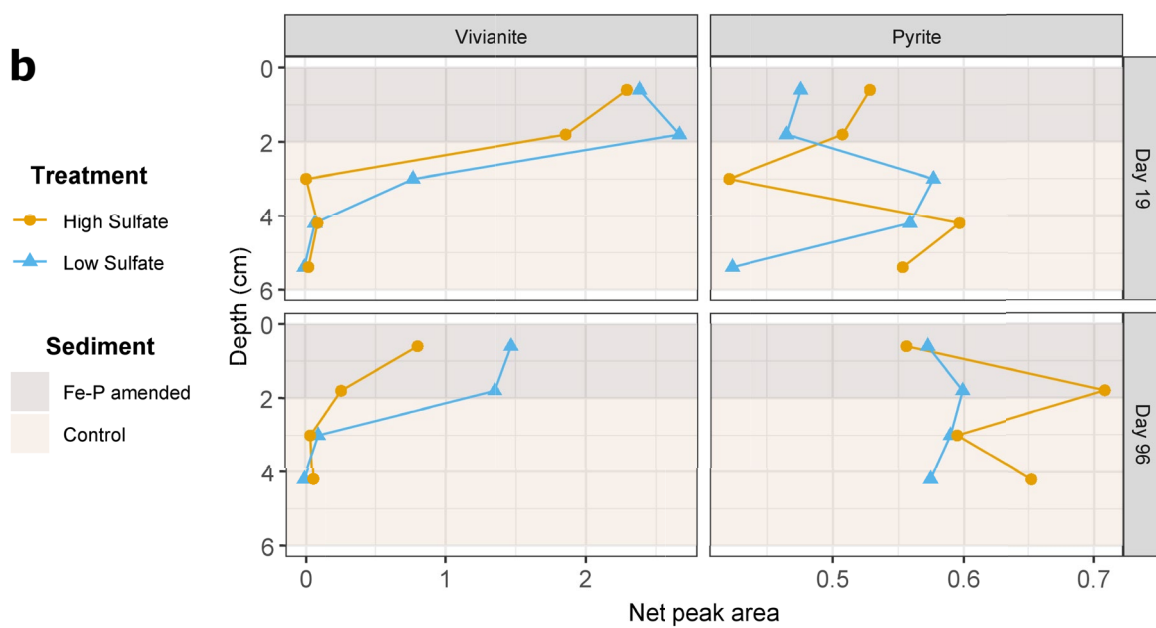


Fig. 3 Porewater conditions in the incubated sediment. **a** Oxygen gradient in the top centimeter of sediment. The measurements are normalized by the upper and lower values: 100% indicates oxygen saturated water under local conditions in the water column achieved by aeration, and 0% the anoxic conditions of the deep sediment. **b** Total dissolved sulfide (TS) concentrations as function of depth, at different times after the start of the treatment, derived from H_2S and pH measurements.

In the high sulfate treatment porewater sulfide increased over time, in contrast to the low sulfate treatment. **c** Iron(II) captured in the DGT binding layer at the end of the experiment. The surface concentration is a function of diffusion from local porewater and Fe(II) release from the solid phase. **d** Phosphorus bound in the DGT at different timepoints of the sulfate incubation



b



◀**Fig. 4** **a** Typical XRD pattern for the Fe-P amended sediment. The main crystalline phases were identified by comparing to reference patterns of the crystallography open database, which are marked by color. While calcite and quartz peak intensity was similar across all sediments, vivianite reflections varied depending on treatment, maximal in the Fe-P amended sediment and absent in the untreated sediment. **b** Relative peak intensity of selected reflections characteristic for vivianite and pyrite. A characteristic reflection was chosen to isolate relative changes specific to that mineral, highlighted by an arrow in the diffractogram above. For Vivianite this was the peak between a 2θ of 14.8° and 15.7° , and for pyrite the peak between 47.45° and 48°

= 0.012). Zooming in on specific sediment layers (Fig. S3) revealed a difference between treatments found in the 1.2–2.4 cm layer (ANOVA, $p = 0.024$), where lower P concentrations were found in the high sulfate treatment compared to the low sulfate treatment. The average P concentration decreased between day 19 and day 96 from 132 to $116 \mu\text{mol P g}^{-1} \text{DW}$ in the low sulfate and from 121 to $89 \mu\text{mol P g}^{-1} \text{DW}$ in the high sulfate treatments. At the end of the experiment, the top slices of the low and high sulfate treatments – which contained the bulk of the P in the amended sediment – had respectively 15–46 % and 36–67 % lower levels of P with than the amended sediment before the start of the incubation ($245 \mu\text{mol P g}^{-1} \text{DW}$).

Sulfur (S) showed the largest compositional change between the start, day 19 and day 96: Average S content was higher in the high sulfate treatment compared to the low sulfate treatment (ANOVA, $p = 0.041$). Compared to the starting composition of $590 \mu\text{mol S g}^{-1} \text{DW}$ the S content in the high sulfate treatment was 6% higher after 19 days ($628 \mu\text{mol S g}^{-1} \text{DW}$) and a further 7% more at day 96 ($674 \mu\text{mol S g}^{-1} \text{DW}$), a total 14 % increase from the start to end. This enrichment was mainly found in the top two depth intervals: The 0–1.2 cm and 1.2–2.4 cm slices contained respectively 17 % and 12% more S at day 96 (687 ± 24 and $720 \pm 4 \mu\text{mol S g}^{-1} \text{DW}$) than the same depths at day 19 (589 ± 25 and $645 \pm 26 \mu\text{mol S g}^{-1} \text{DW}$). The low sulfate treatment, by contrast, remained unchanged with respect to S content.

Iron, phosphorus and sulfur mineral-phase speciation

XRD identified major crystalline phases in all samples, including quartz, calcite, and the Fe-sulfide

mineral pyrite (FeS_2). The pyrite signal increased over time, by 10 % and 20 %, in the low and high sulfate treatment, respectively (Fig. 4b). The characteristic vivianite reflection pattern (Fig. 4a) was present in all Fe-P amended sediment but absent in the control. By day 96, the vivianite XRD signal in the top 2 cm of the high sulfate treatment was 70–90 % lower than in the pre-incubation amended sediment and 65–85 % lower than at day 19; in the low sulfate treatment, the corresponding reductions were 40–45 % and 38–49 %, respectively (Fig. 4b).

SEM-EDS confirmed these observations: Phase identification maps showed Fe-sulfide phases distributed evenly as sub-micrometer particles, some displaying a framboidal shape typical of sedimentary pyrite (Rickard 2012), with an EDS S:Fe of approximately 2:1 (Table 2). Vivianite crystals were identified by their characteristic tabular shape in SE images (Fig. 5b), with an EDS Fe:P of 3:2 (Table 2). Associated Fe-rich phases, most likely FeOx, showed higher Fe:P approaching 4:1 and appeared as amorphous $\sim 10 \mu\text{m}$ clusters (Fig. 5b,c) typical of ferrihydrite, the main constituent of the amendment material (Schöpke et al. 2013). In addition to residual ferrihydrite, freshly precipitated FeOx was visible as orange-brown deposits in the upper sediment and on the DGT gel (Fig. 2c).

Changes in P and S pools

Phosphorus was extracted mainly in the BD-P and NaOH-SRP fractions (Fig. 6). The NaOH-SRP pool declined between day 19 and day 96 in both treatments (ANOVA, $p = 0.00092$), although the low sulfate treatment retained more NaOH-SRP than the high sulfate treatment (ANOVA, $p = 0.0015$). The average NaOH-SRP concentration following the high sulfate treatment was 70% lower at day 96 ($32 \mu\text{mol P g}^{-1} \text{DW}$) than at day 19 ($9 \mu\text{mol P g}^{-1} \text{DW}$), and 40% lower for the low sulfate treatment ($40 \mu\text{mol P g}^{-1} \text{DW}$ to $19 \mu\text{mol P g}^{-1} \text{DW}$). Depending on the depth, the NaOH-SRP pool lost up to $60 \mu\text{mol P g}^{-1} \text{DW}$ and decreased by 28–90% in the high sulfate treatment and 48–76% in the low sulfate treatment, most in the top layers. The BD-P pool on average remained stable over time (ANOVA, $p = 0.44$) but did decrease by approximately 25% in the sulfidic zone (1.2–2.4 cm) of the high sulfate treatment. The BD-Fe

Table 2 Phases identified by EDS-spectroscopy in the dried sediment. The Fe-P amended sediment before the start of the sulfate treatment was investigated, as well as sediment slices from 1.2–2.4 cm depth after different periods of both low and high sulfate treatment. Electron microscope images and EDS mapping of the samples are shown in Fig. 5, with the identified Fe phases highlighted. The relative abundance of detectable elements was determined by analysis of the combined spec-

trum of each total area, and specific areas containing one or more phases of interest. The EDS spectrum of a specific phase was determined by either the automatic phase detection algorithm in the EDAX software or by manually selecting a rectangular area containing an identified phase. The automatic phase detection also reported the percentage of pixels assigned to the phase on the map, as measure of the surface concentration

Phase	Treatment time (days)	Sample treatment	Stoichiometry (mol %)			Surface Conc. (area%)	Phase selection method
			Fe	P	S		
Vivianite	0	Fe-P amendment	n.a.	n.a.	n.a.	8	manual
	19	Low sulfate	9.0	6.7	0.2	2	automatic
		High sulfate	9.0	6.0	0.0		manual
Iron sulfide	0	Fe-P amendment	3.3	0.2	6.0	2	automatic
	19	Low sulfate	10.0	0.3	21.2	1	automatic
		High sulfate	4.0	0.1	7.6	1	automatic
		Low sulfate	7.7	0.1	13.1	1	automatic
		High sulfate	4.6	0.1	9.0	4	automatic
Iron hydroxide	19	Low sulfate	22.5	5.9	1.0	2	automatic
		High sulfate	9.0	2.3	0.6	3	automatic
Selected area	19	Low sulfate	4.6	2.3	1.4		
Total area	0	Fe-P amendment	0.8	0.2	0.9		
	19	Low sulfate	1.3	0.4	1.0		
		High sulfate	0.7	0.0	0.5		
		Low sulfate	0.8	0.1	1.2		
		High sulfate	0.9	0.1	1.6		

concentration increased by 35% in the top layer of both treatments relative to the starting material.

A dark band of sulfide mineral precipitation appeared within a few days at approximately 2 cm depth in all treatments, at the interface between the Fe-P amended and the underlying sulfidic sediment and became visibly darker in the high sulfate treatment over time (Fig. 2). This was corroborated by sulfide extraction results: extractable S was enriched in the Fe-P amended layer (0.5–2.5 cm depth) relative to the lower sediment in both treatments after 96 days (Fig. 7a). Total extracted S (S_{extr} : AVS + CRS) at 0.5–2.5 cm depth was 90–140% and 40–60% higher than in the lowest layers in the high and low sulfate treatments, respectively. This S enrichment increased the molar $S_{\text{extr}}:\text{Fe}_{\text{ICP}}$, reflecting the formation of FeS_x minerals with varying stoichiometries, reaching up to 2.5 in the high sulfate treatment — above the maximum S:Fe of 2 for Fe-sulfides, suggesting the presence of additional non-Fe-sulfide S phases such as S^0 .

The lowest values (~1) were found near the surface of both treatments, higher than the pre-treatment ratio (1.4), indicating a likely increase in non-sulfidic Fe.

Comparing S enrichment between treatments and fractions as the concentration difference between the most altered slice (2 cm) and the least affected slice (4 cm): sulfur was predominantly in the CRS fraction and partly in AVS. Relative enrichment was greater in the AVS fraction (826% and 263% for high and low sulfate treatments, respectively) than in the CRS fraction (59% and 44%). However, the absolute enrichment was larger in the CRS fraction (up to 256 $\mu\text{mol S g}^{-1}$ DW) than in the AVS fraction (max. 153 $\mu\text{mol S g}^{-1}$ DW).

Discussion

This experiment was designed to trace the fate of iron-bound P in Fe-amended lake sediment under

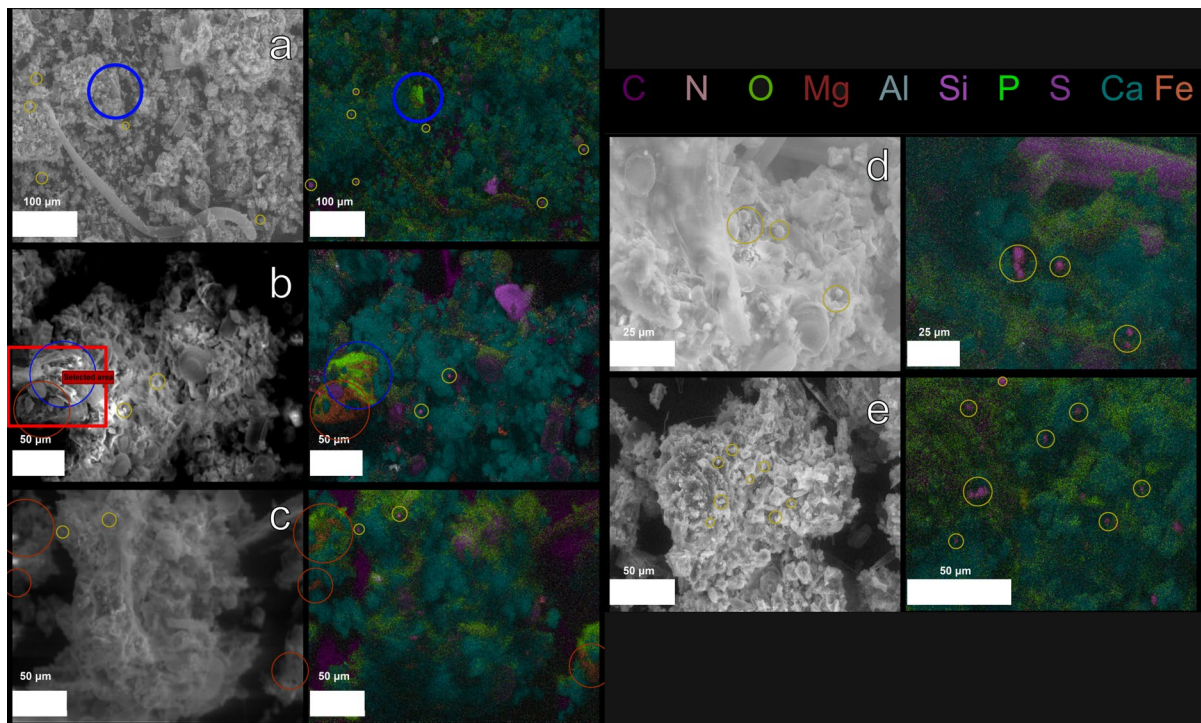


Fig. 5 Scanning electron microscopy and energy dispersive spectroscopy results of selected sediment samples. **a** The Fe-P amended sediment after pre-incubation but before the sulfate treatment. **b**) and **c**) samples from 1.2–2.4 cm sediment depth of day 19 of the low and high sulfate treatment, respectively. **d**) and **e**) Sediment from this depth of respectively the low and high sulfate treatment at the end of the experiment (day 96). Left: SEM Secondary Electron images of a grain of dried sediment, showing the sample topography. Right: Elemental distri-

bution within the same area by EDS mapping, each pixel color corresponds to the element with the highest normalized intensity, which shows the location of certain phases rich in specific elements. These phases were identified and are highlighted with circles: Blue) Vivianite; Red) Iron hydroxide; Yellow) Iron sulfide. A selected area which contains multiple phosphorus binding iron phases is also highlighted. The stoichiometry of Fe, P and S of the identified phases and selected area can be found in Table 2

defined sulfidic conditions. The central pattern is a decline in solid-phase P over the 96-day incubation, while S accumulated under high sulfate conditions. Changes in P retention were greater under high sulfate than under low sulfate, highlighting the important role of sulfur on the persistence Fe-bound P. The unique setup of this experiment provided a well-defined redox gradient, which revealed that the compositional shifts were most pronounced at depth where sulfide production was greatest.

Redox conditions in the sediment

During the incubation, the sediment developed visibly separated redox zones (Fig. 2), supported by pore-water data. Oxygen microprofiles indicated that the

oxic zone was confined to the top <0.3 cm (Fig. 3a). Below this, the sediment was anoxic, with evidence of anaerobic microbial respiration — including both Fe(III) and sulfate reduction. Reductive dissolution of mineral-oxides decreased solid-phase Fe and Mn content at 1.2–2 cm depth, while DGT data showed elevated porewater Fe(II) peaking at 1.5 cm in the low sulfate treatment and 2 cm in the high sulfate treatment. The curvature of TS profiles between 2 and 3 cm in the high sulfate treatment indicated the onset of a sulfate reduction zone at those depths, consistent with the canonical layering of oxic, iron/manganese-reducing, and sulfate-reducing zones (Grundl et al. 2011). Higher porewater TS in the high sulfate treatment is consistent with increased sulfate reduction activity at higher sulfate availability (Holmer and Storkholm 2001), and were within the ranges reported

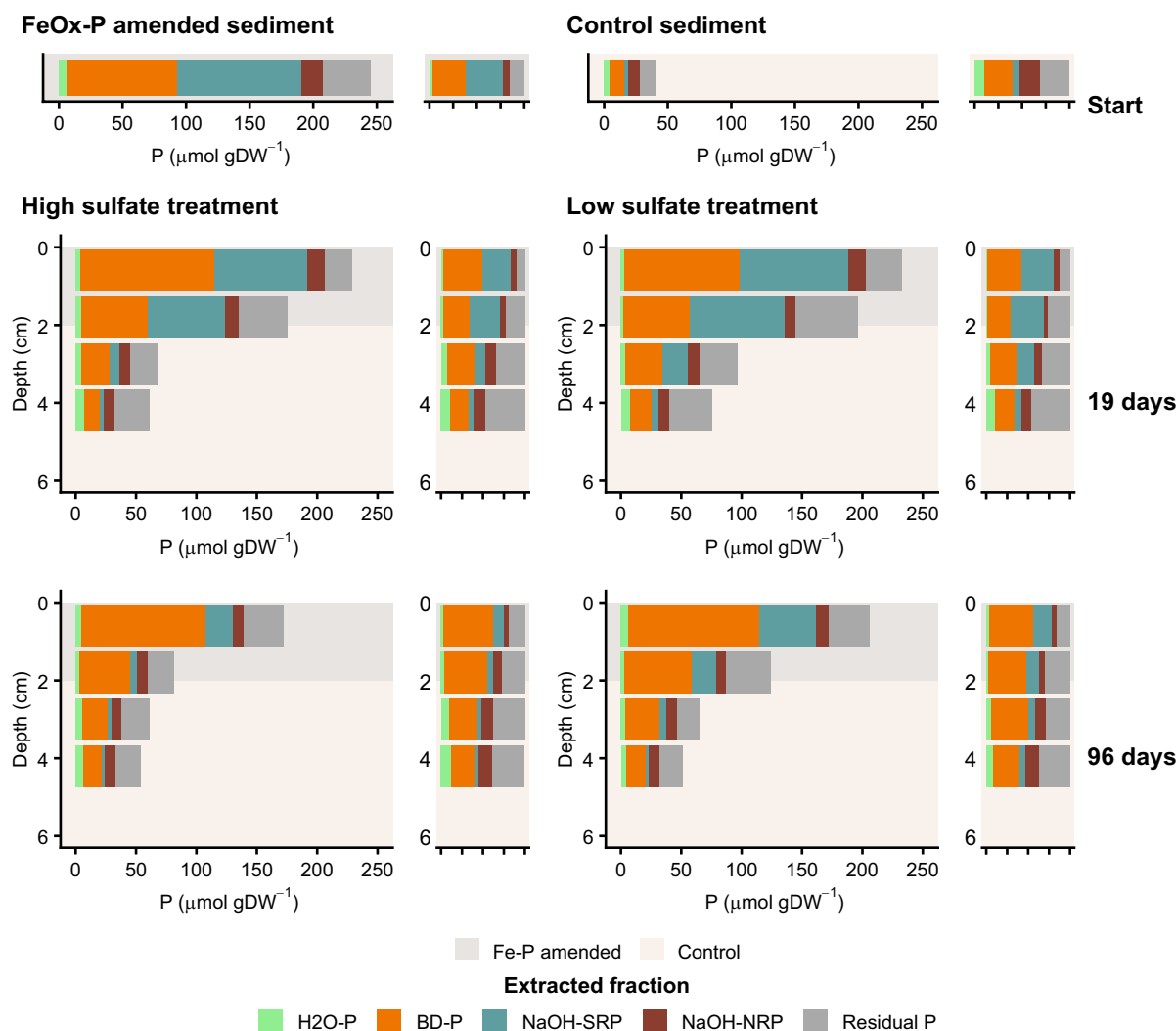


Fig. 6 Extracted P fractions in the Fe-P amended and unamended control sediment after pre-incubation, sampled before the start of the sulfate treatment (top row), and after respectively 19 and 96 days (below) shown as function of depth. The amount extracted P per gram of dry sediment is shown in the left side of each window, while the right side shows the same

data normalized, to highlight the relative contribution of each fraction to the total P. The sequential extractions target different P forms in the sediment: H₂O-P: loosely bound P; BD-P: redox-sensitive P; NaOH-SRP: metal-bound P; NaOH-NRP: organic-bound P; residual P: not extracted P forms i.e. apatite

from previous studies in freshwater lakes (Sandfeld et al. 2020; Steininger et al. 2023).

The influence of the oxic surface layer on Fe and P cycling was evident: the P content remained more stable in the oxic top sediment than in the anoxic sediment below. Diffusive gradient in thinfilm profiles suggest that Fe(II) released by FeOx reduction diffused upward into the oxic zone, where it was oxidized and reprecipitated as visible orange-brown deposits on the sediment surface and in the DGT gel

(Fig. 2c). Because the top sediment slice extended well beyond the ~0.5 cm oxic zone, bulk Fe content in this slice did not change over time, as FeOx accumulation was likely offset by Fe reduction in the anoxic portion of the slice. The capacity of this freshly precipitated FeOx to bind P likely explains the elevated BD-P and BD-Fe in the uppermost layer — a cycle of reoxidation and P adsorption previously described as a mechanism for continued P retention after Fe amendment (Kleeberg et al. 2013; Busigny et al.

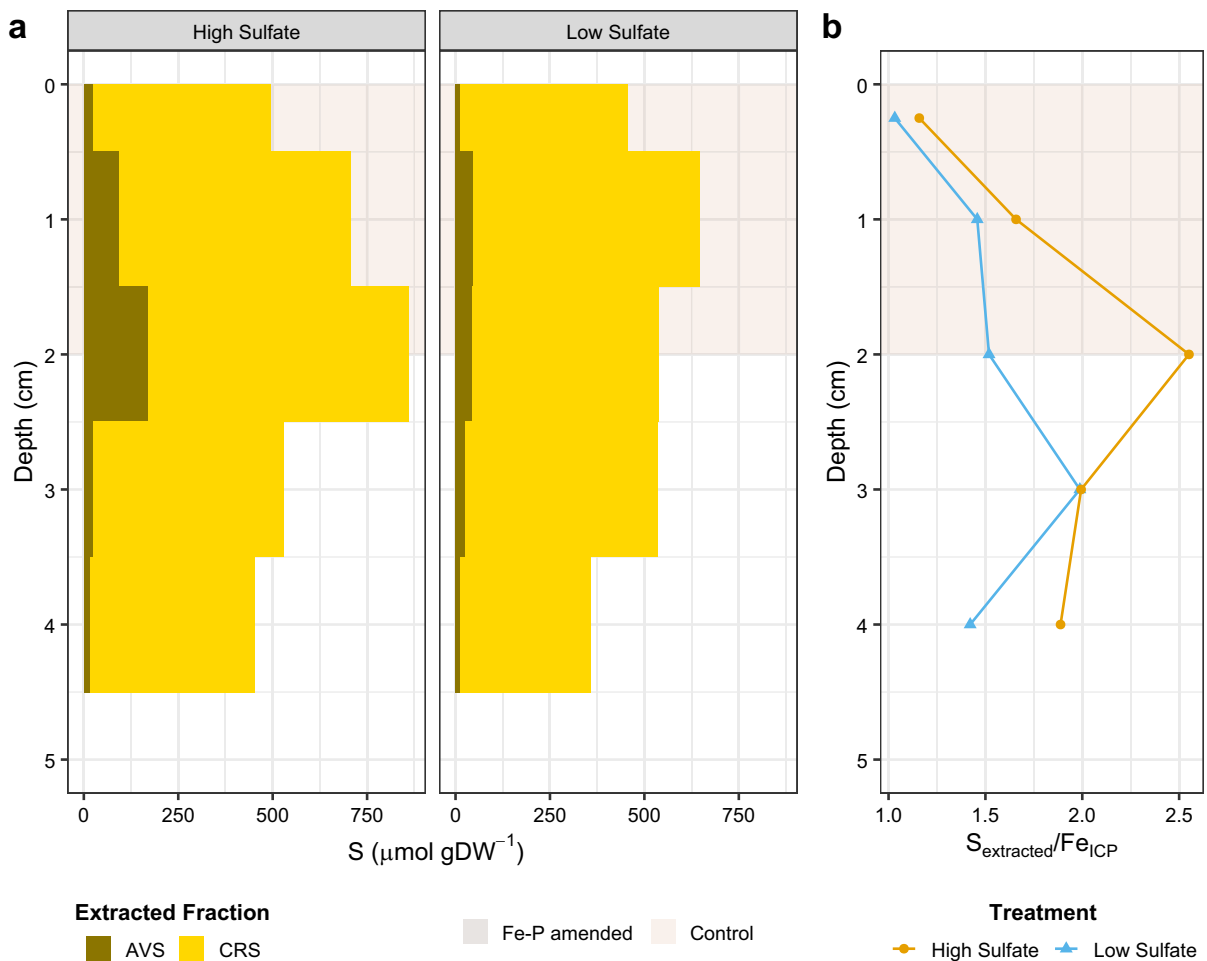


Fig. 7 Sulfur extraction results. **a** Sulfide extracted in the acid volatile (AVS) and chromium reducible (CRS) fractions, as function of depth in the sediment after 96 days of incubation. The sediment is enriched in sulfide in the zone with Fe-P amended sediment (0.5–2 cm), and highest in the high sulfate

treatment. While the enrichment is partly found in the AVS fraction, the largest increase is found in the CRS fraction. **b** The molar ratio between the combined sulfide fractions (AVS + CRS), and the measured solid phase Fe at the same depth

2016). Nevertheless, total solid-phase P decreased over time in both treatments, indicating net P mobilization despite this FeOx trap.

Phosphorus mobilization processes

The decrease in solid-phase P was most pronounced in the sulfidic zone of the high sulfate treatment, consistent with the role of sulfide in P mobilization from Fe-P phases (Kleeberg 1998; Heinrich et al. 2022). The substantial decline in NaOH-SRP identifies vivianite dissolution as the main source of mobilized P,

supported by XRD data. These results align with our hypothesis that sulfide levels strongly control vivianite stability. Nonetheless, even the most affected sediment slices still contained some vivianite, indicating that persistence might also depend on the magnitude/duration of sulfide exposure as well as mineral properties like crystal surface area.

Phosphorus content also declined over time in the sediment under low sulfate treatment, where net sulfide accumulation was below 50 μmol L⁻¹, suggesting additional P release mechanisms. Since Fe-P was only partially transformed to vivianite during pre-incubation, direct microbial reduction of FeOx-P

likely continued in the anoxic sediment, driving migration of the BD-P pool toward the oxic zone. Sulfide may also have played a role even in the low sulfate treatment, as H_2S can in addition to sulfate reduction be liberated from OM mineralization (Dunnette et al. 1985). The tight coupling between H_2S generation and its rapid re-oxidation (Sandfeld et al. 2020), and/or precipitation with Fe from P-bearing phases may explain the decrease of detectable free sulfide in the porewater of the low sulfate treatment, even at the expected sulfate reduction depth (1.5–2.5 cm).

The decline in the redox sensitive BD-P fraction in the sulfidic zone of the high sulfate treatment, but not in the low sulfate treatment, indicates that FeOx-P dissolution is also sensitive to sulfide levels, though its contribution to P release was smaller than that of vivianite. This difference in contribution is partly explained by reaction stoichiometry: The reaction of FeOx with sulfide involves reductive dissolution of Fe(III) (Bronner et al. 2023) consuming additional sulfide compared to the direct reaction with Fe(II) in vivianite, which already holds more P per mole Fe than FeOx (Fe:P of vivianite: 1.5; for FeOx typically below ~10 but rarely below 2:1; Gunnars et al. 2002). Consequently, approximately 10 times more P is released per mole of sulfide for vivianite than for FeOx.

Although Fe(II) in vivianite is theoretically vulnerable to oxidation, this does not necessarily lead to dissolution; instead, an amorphous Fe(III) surface layer can form, protecting against further dissolution (Rouzies and Millet 1993; Rothe et al. 2014; Metz et al. 2024). Our results similarly show vivianite persisting in the oxic layer, still detectable by XRD after ~100 days (Fig. 4). The long-term stability of vivianite under oxic conditions depends on crystal size, ligand availability, cation substitution, and natural impurities (Kubeneck et al. 2024; Metz et al. 2024; Banke et al. 2025), and using sediment-produced rather than synthetic vivianite in our experiment is likely more representative of naturally occurring mineral reactivity.

Solid-phase sulfide accumulation in the sediment

Sulfide reacted with Fe(II) to form FeS_x minerals, visible as characteristic black precipitation in the

sediment and on the DGT gel at 1.5–2.5 cm depth. This immobilized dissolved Fe(II), resulting in lower porewater Fe(II) levels in the high sulfate cores, where net Fe(II) accumulation in the gel was reduced by two-thirds compared to the low sulfate treatment. Simultaneously, sulfide reacting with either dissolved Fe(II) or solid-phase Fe(III) also immobilized S. The absence of porewater TS in the uppermost centimeters suggests that upward sulfide diffusion was effectively prevented by reaction with Fe, though this buffering capacity appears to diminish progressively over the incubation, leading to an upward shift of the Fe-P precipitation horizon, observed as increased coloration of the sediment (Fig. 1a) and higher concentrations of sulfide at shallower depths (0.5–1 cm).

Precipitated Fe-sulfide phases were distributed across both the AVS and CRS fractions (Fig. 7). The CRS fraction targets the pyrite pool (Hsieh and Shieh 1997), and the 10–40% increase in the crystalline pyrite XRD signal over time and between treatments provides evidence of pyrite growth (Rickard 2012), although the increase in pyrite signal correlates only weakly with the CRS fraction between samples at the end of the experiment ($r = 0.44$). The substantially larger CRS enrichment (30–100%) relative to the crystalline pyrite increase may be explained by the production of S^0 from FeOx reductive dissolution with H_2S , which is also extracted in the CRS fraction (Bronner et al. 2023). The large AVS enrichment in the reactive zone (0.5–2.5 cm) of the high sulfate treatment reflects the formation of reactive iron monosulfide (FeS, mackinawite) — consistent with the established pathway in which dissolved Fe(II) and sulfide first form amorphous FeS, which subsequently reacts with additional sulfide to form pyrite (Rickard and Luther 2007; Rickard 2012; Kraal et al. 2013).

Representativeness for real systems

The setup of this experiment was explicitly artificial, optimized for investigation of the sediment processes, and this simulated system cannot be directly transferred to a real lake context. The stoichiometric inferences drawn here, like the coupling of sulfide production to vivianite dissolution, the partitioning of iron and the fate of the liberated phosphorus required that all incubation units begin from an identical and known solid-phase composition, hence the use of

composed upper sediment from Langer See. The consequence is that the experiment removes the natural sample-to-sample variability of the lake by design. The vivianite-enriched sediment should likewise be regarded as a model system rather than a representative one. Its reactive Fe content lies at the upper end of the range reported for iron-amended lake sediments (Bakker et al. 2016), while sediments with a higher ratio of reactive iron to sulfide production should be more resilient than the system studied here.

The quantitative outcomes reported here are therefore properties of this system under these imposed conditions, and they should not be transferred to the field as expected values. A real Fe-amended lake can be investigated by analyzing undisturbed cores (Münch et al. 2024), but in general, a mesocosm approach alone cannot establish whether the effect of sulfate after Fe amendment would be detectable at the scale of the lake, testing this would require a combined *in-situ* study.

The mechanistic argument does still hold for real systems: The sequence we observe — microbial sulfate reduction, sulfide accumulation, dissolution of vivianite, precipitation of amorphous Fe sulfides and pyrite, release of the associated phosphorus, and its partial re-adsorption onto freshly precipitated Fe(III) phases at the oxic sediment surface — is governed by the relative stabilities and formation kinetics of the phases involved more than by the spatial variance of the starting material. We therefore expect the direction of this pathway to hold in intact sediment. The rates and the endpoint, by contrast, will depend on the local reactive Fe:S:P stoichiometry, on the supply of OM as an electron donor for sulfate reduction, and on bioturbation and bioirrigation, all of which were not covered in the present study.

Considerations for Fe amendment practices

The outcomes of Fe-based lake restoration are notably variable (Boers et al. 1994; Deppe and Benndorf 2002; Kleeberg et al. 2012; Heinrich et al. 2022; Münch et al. 2024). Knowledge of lake-specific conditions is essential for predicting the outcome and timescale of Fe amendments, and the measure is only effective when integrated into a system-wide approach that also mitigates external loading (Lürling et al. 2016). While sediment

composition provides useful initial information and attempts have been made to prescribe Fe dosage based on local P, Fe, S, and OM chemistry (Geurts et al. 2008; Kleeberg et al. 2012), stability assessments based solely on bulk sediment characteristics are insufficient. Fe reactivity and the influence of destabilizing factors evolve during sediment maturation, so ongoing monitoring of P retention capacity is necessary. Ideally, assessment of Fe-P stability should detect incipient internal P loading before it leads to a return of eutrophic conditions, allowing timely application of additional measures such as Fe re-dosing or mitigation of sulfate loading.

Heinrich et al. (2022) proposed assessing the sulfate reduction rate to account for sulfide competing with P for Fe binding. However, the sulfate reduction rate is not trivial to measure and varies widely over time; moreover, while sulfate reduction generates sulfide (that can largely be reoxidized, e.g. Sandfeld et al. 2020), it is not itself a direct P-releasing process. The extent to which sulfide production translates into P release depends on diffusion, kinetics, and reaction stoichiometry. Unlike prior studies that focused on the thermodynamic instability of Fe-P in the presence of sulfide, our study directly measured absolute sulfide concentrations at the depths where Fe-P and Fe-S reactions occur within a model system. This provides a tangible parameter linking local sulfide concentrations to vivianite destabilization and Fe sulfide mineral formation: sulfide reacting with Fe lowers porewater sulfide concentrations, which in turn controls reaction rates (Poulton et al. 2004), until a quasi-steady state is reached between sulfide production and consumption. In our experiment porewater sulfide is measurable at high vertical resolution and rises before phosphorus reaches the water column, which merits evaluation as an early indicator that iron-based phosphorus retention is failing. To monitor P retention, reaction rates and mass balance could ultimately be evaluated using kinetic models, adjusted to local conditions, and reliant on measured sulfide concentrations in treated lakes (Wu and Wang 2017). Recent advancement in sensing technology allow to measure total dissolved sulfides at high spatial and temporal resolution without the need of simultaneous pH measurements (Steininger et al. 2023) providing an excellent occasion for rapid assessment and to qualify models.

Conclusions

Our results demonstrate that vivianite stability is highly sensitive to sulfidic conditions. Sulfide produced by microbial sulfate reduction triggers coupled dissolution–precipitation reactions that transform vivianite-bound Fe into amorphous Fe sulfides and pyrite, releasing 60–90% of vivianite-bound P in the high sulfate treatment within the 96-day experiment. This demonstrates that sulfide production reduces the capacity of the sediment to bind P in a stable form and can lead to P release on a timescale of months — a period relevant for natural lake systems. Although the contributions of individual P-releasing processes were not fully quantified, vivianite dissolution clearly represented the dominant source of mobilized P in this system.

Our study directly links local porewater sulfide concentrations to specific mineral transformations at the depths where Fe–P and Fe–S reactions occur. This mechanistic framework reveals that multiple processes — reductive FeOx dissolution, vivianite destabilization, and Fe sulfide mineral formation — act simultaneously and can be related to measurable sulfide dynamics. This study was conducted on a model system, and although the magnitude of the here described process may vary locally, due to heterogeneity of organic matter distribution and from site to site, our data indicate that monitoring porewater sulfide concentrations can serve as a practical early warning indicator of declining P retention capacity following Fe amendment. These insights align with and extend previous frameworks proposing that effective Fe dosage calculations must account for the sediment sulfur pools.

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Author contributions The study was designed by Harm van Kuppevelt, Ugo Marzocchi and Michael Hupfer, with assistance of Kasper Reitzel. The experiment was performed by Harm van Kuppevelt, with help of Ugo Marzocchi. Sulfide extraction measurements and analysis were done by Marta Lidia Sudo. Further data gathering and evaluation was done by Harm van Kuppevelt. The manuscript was drafted by Harm van Kuppevelt who also revised the manuscript after commentary from Michael Hupfer, Ugo Marzocchi and Kasper Reitzel. All authors read and approved the final manuscript.

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Data availability The data analysed in this study are deposited in the Zenodo repository (<https://doi.org/10.5281/zenodo.22147812>). The raw data can be obtained from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no relevant financial or non-financial interests.

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